Blood Bond Portal

Complete Setup, Installation & Execution Guide (Step-by-Step)

TECHNICAL DOCUMENTATION

1. Prerequisites & Official Software Downloads

Ensure the following three core software packages are downloaded and installed on your host system prior to launching the project:

Software Application	Official Download URL	Purpose & Environment
Node.js (LTS Version)	`https://nodejs.org/`	Backend server runtime (v18.x or v20.x recommended)
MySQL Server & Workbench	`https://dev.mysql.com/downloads/installer/`	Relational database storage & administration tool
Visual Studio Code	`https://code.visualstudio.com/`	Integrated development environment & integrated terminal

2. Software Installation & Environment Setup

A. Installing Node.js

1. Download the **LTS (Recommended For Most Users)** Windows installer (`.msi`) from the official portal.
2. Launch the installer and proceed with **Next > Accept License Agreement > Next > Install**.
3. Once completed, open Command Prompt or PowerShell and verify installation via `node -v` and `npm -v` to verify proper environment path configuration.

B. Installing MySQL Server & Workbench

1. Download the MySQL Community Installer from the official developer repository.
2. Select the "**Developer Default**" configuration or choose custom installation containing **MySQL Server** and **MySQL Workbench**.
3. During configuration, establish a secure **MySQL Root Password** (e.g., `root` or a custom password of choice). Retain this credential for the project environment file.
4. Proceed through the configuration steps and click **Execute** to complete service initialization.

3. Project Workspace & Database Initialization

Step 1: Extracting the Project Archive

1. Locate the downloaded project archive package: **`blood_bond.zip`**.
2. Right-click the compressed archive, select "**Extract All...**", and designate an accessible directory (e.g., `Documents` or `Desktop`).
3. Launch the extracted folder directly in **VS Code** (Right-click folder > Open with Code).

Step 2: Restoring the Database via MySQL Workbench

1. Open **MySQL Workbench** and authenticate into your **Local instance (root)** connection.
2. Navigate to the top toolbar: select **File > Open SQL Script...**
3. Browse to the project root directory and select the database schema file: **`schema.sql`**.
4. Click the execute icon (⚡ **Yellow Flash Icon**) in the SQL editor toolbar to create the database schema and requisite entity tables.

4. Environment Configuration & Server Execution

Step 3: Configuring the Environment Variables (`.env`)

Open the `.env` file located in the project root directory and ensure database connection parameters correspond to your local MySQL instance:

```
DB_HOST=localhost DB_USER=root DB_PASSWORD=YOUR_MYSQL_PASSWORD_HERE DB_NAME=blo
od_bond PORT=3000 EMAIL_USER=shamchavan4444@gmail.com EMAIL_PASS=yptpuvsdtmduzsqa
```

Step 4: Installing Dependencies & Starting Application

1. Open the integrated terminal within VS Code: **Terminal > New Terminal** (or press `Ctrl + ``).
2. Execute the package installation command to pull all required node modules:

```
npm install
```

3. Launch the application server with:

```
node server.js
```

4. Observe the terminal console confirming: **Blood Bond server listening at http://localhost:3000**.
5. Open any web browser and navigate to: **`http://localhost:3000`**

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